DAY 03 ASSIGNMENT
NETWORK INTERFACES AND CABLES

Submitted By
SREEKAKULAM THULASIRAM

1) THEORY

1. What is a Network Interface Card?

A Network Interface Card is a piece of hardware which enables any device to connect to a network. They can support both wired and wireless connections. Their role is to enable devices to send and receive data within the network. Some computers now have Network Interfaces embedded into the motherboard.

2. What is Ethernet?

Ethernet is a widely used technology which enables devices to connect within a Local Area Network. Mostly uses cables to send data between computers, switches and routers.

3. What is RJ-45?

An widely used term for an 8 pin modular jack. Is used for Ethernet network cables. Connects Ethernet cables to computing devices like computers, switches and routers. Has 8 pin contacts to send data along the network.

4. What does UTP stand for?

UTP stands for Unshielded Twisted Pair. This is a type of network cable which contains insulated copper pairs. The wires are twisted together to reduce electromagnetic interference.

5. Name 3 common Ethernet cable types.

Cat5e (Category 5e) Cat6 (Category 6) Cat6a (Category 6a)

6. What is a straight through cable?

Straight through cables are Ethernet cables where both ends are wired the same, either T568A or T568B. They are commonly used to connect different types of devices together.

7. What is a crossover cable?

A crossover cable is an ethernet cable were one end is wired using T568A and the other end T568B. Traditionally used to connect similar devices together.

8. Explain what Auto MDI-X means.

Auto MDI-X means Automatic Medium Dependent Interface Crossover. Many modern ethernet devices have this functionality. This means when you connect a cable it will automatically determine what type of cable is needed and swap the transmit and receive pairs so straight through and crossover cables can be used.

9. What are the differences between copper and fibre optics?

Copper networking uses electrical signals to send data along the copper wire. Commonly used in Ethernet LANs. Generally cheaper to install.

Fibre Optic networking uses light signals to send data along thin glass/plastic fibre. Offers high bandwidth and longer distances for data transmission. Not affected by electromagnetic interference.

10. What is multimode fibre?

Multimode fibre is a type of fibre optics allowing more than one light path/mode to go through the core. Generally used for short distances, such as in building and data centres. Commonly uses LED or VCSEL light sources depending on what system it is.

11. What is single mode fibre?

Single mode fibre is a type of fibre optic with a very small core allowing light to travel mostly in one mode. Used for longer distances and high bandwidth networks. Commonly uses laser light sources.

12. 2 benefits of fibre optics.

High bandwidth – fibre optic cables can carry high amounts of data.

No electromagnetic interference – fibre optic cables don’t use electricity so they aren’t affected by electromagnetic interference.

13. Why is it important to understand Network Interfaces for Cybersecurity?

It is important to understand network interfaces as they connect computers to networks. As a security professional, knowing what network interfaces are will help to identify what network interfaces are being used, monitor traffic going through the network and how devices communicate. This can help to identify unauthorised devices, identify suspicious activity, identify possible attacks and secure systems from unauthorised connections.

2) PRACTICAL IDENTIFICATION

14. Identify your active network adapter.

I used the Windows command prompt to identify what network interfaces I have available. Identified my active network adapter by seeing which network adapter was connected.

Active Network Adapter: Wireless LAN adapter Wi-Fi 2:

Network Adapter Description: Intel(R) Wireless-AC 9560 160MHz

15. Identify your IPv4 address, Default Gateway and Network Adapter Name.

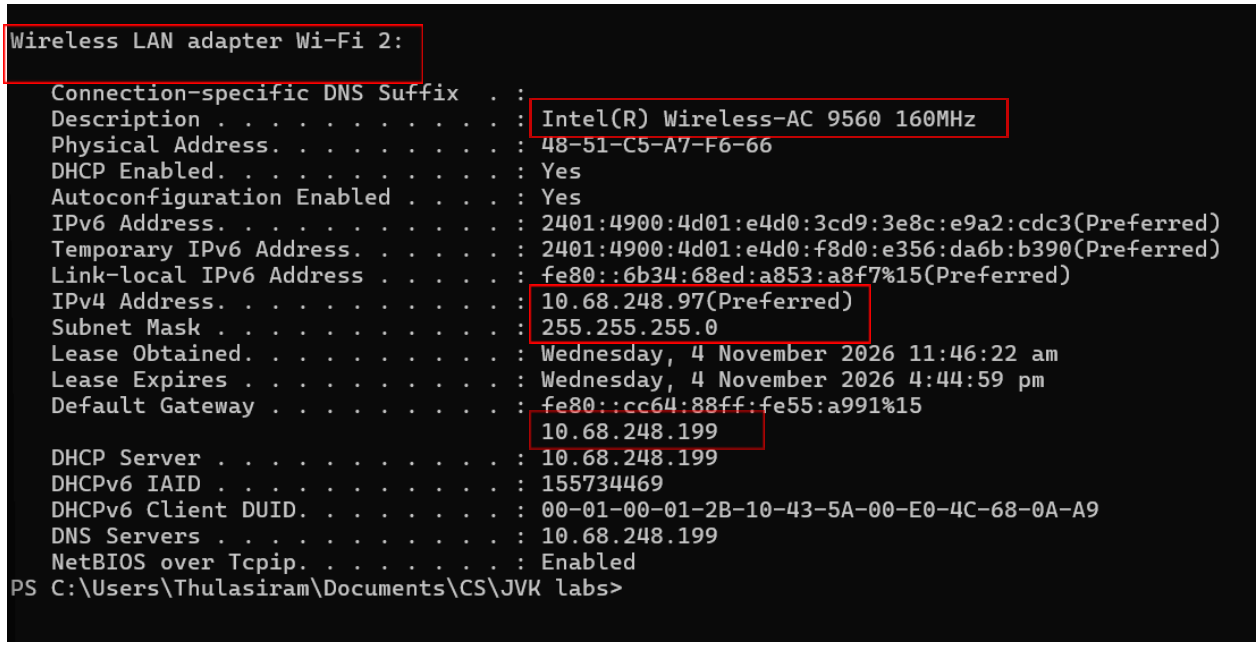
Used the command ipconfig /all to identify my network settings.

Network Information	My Computer Information
Network Adapter Name	Wireless LAN adapter Wi-Fi 2:
IPv4 Address	10.68.248.97
Default Gateway	10.68.248.199

3) SCREENSHOT

Screenshot 1 – IP Configuration

This screenshot shows my network configuration after running the command ipconfig /all in the Windows command prompt. Shows my active network adapter, network adapter description, IPv4 address and Default Gateway.



CONCLUSION

Completing this assignment has taught me about Network Interfaces, Ethernet, RJ-45 and different types of network cables. I have also learnt about the differences between copper and fibre optic networking and the differences between multimode and singlemode fibre. I have learnt how to identify my active network adapters and use the command ipconfig /all to identify my network settings. This is useful to know to understand how network connections work and how security professionals identify and secure network devices.